Superfamily Analysis of Rice Protein Structures Reveals a Variety of Stress Tolerance Mechanisms

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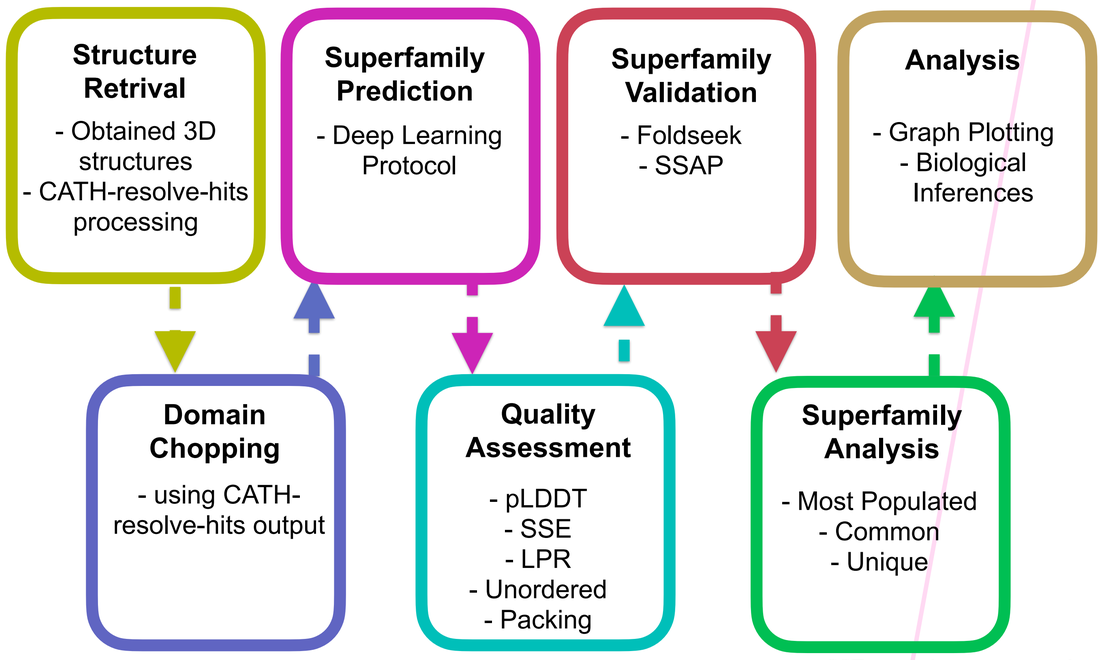
Introduction

Rice crop serves as the staple food for 50% of the global population. However, a large amount of the annual rice yield is lost due to biotic/abiotic stresses affecting rice protein structures[1]. Previously, understanding rice protein structures was difficult due to the low number of experimentally solved structures. Recently, the AlphaFold Protein Structure database[2] has provided models for several organisms, including rice and the model plant, *Arabidopsis thaliana* (Arabidopsis). This leads to several other questions: (1) How's the overall model quality of these models, (2) Can the modelled domains be chopped and mapped into protein domain superfamilies present in CATH database[3]? Hence, this study has been designed to analyse and explore highly populated stress mediating protein superfamilies that are common between rice and Arabidopsis and also those exclusively present in rice.

Methodology

We used all the 43,460 and 27,434 AlphaFold2 3D models belonging to *Oryza sativa* sub sp. *Japonica* (Asian rice), and Arabidopsis, respectively. Almost 99.5% of models belonging to Rice and Arabidopsis were processed using our protocol that yielded 56,621 and 54,593 protein domains. These domains where subjected to various quality assessment steps and validation using CATH SSAP[3] and Foldseek[4] followed by superfamily analysis.

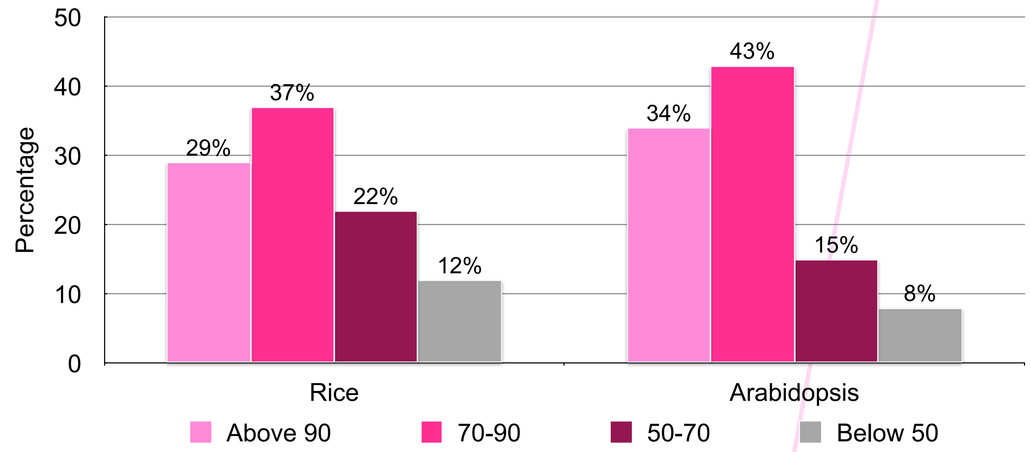


Schematic digram showing the steps followed in a sequential manner

SSE-secondary-structure element
LPR-long problematic region

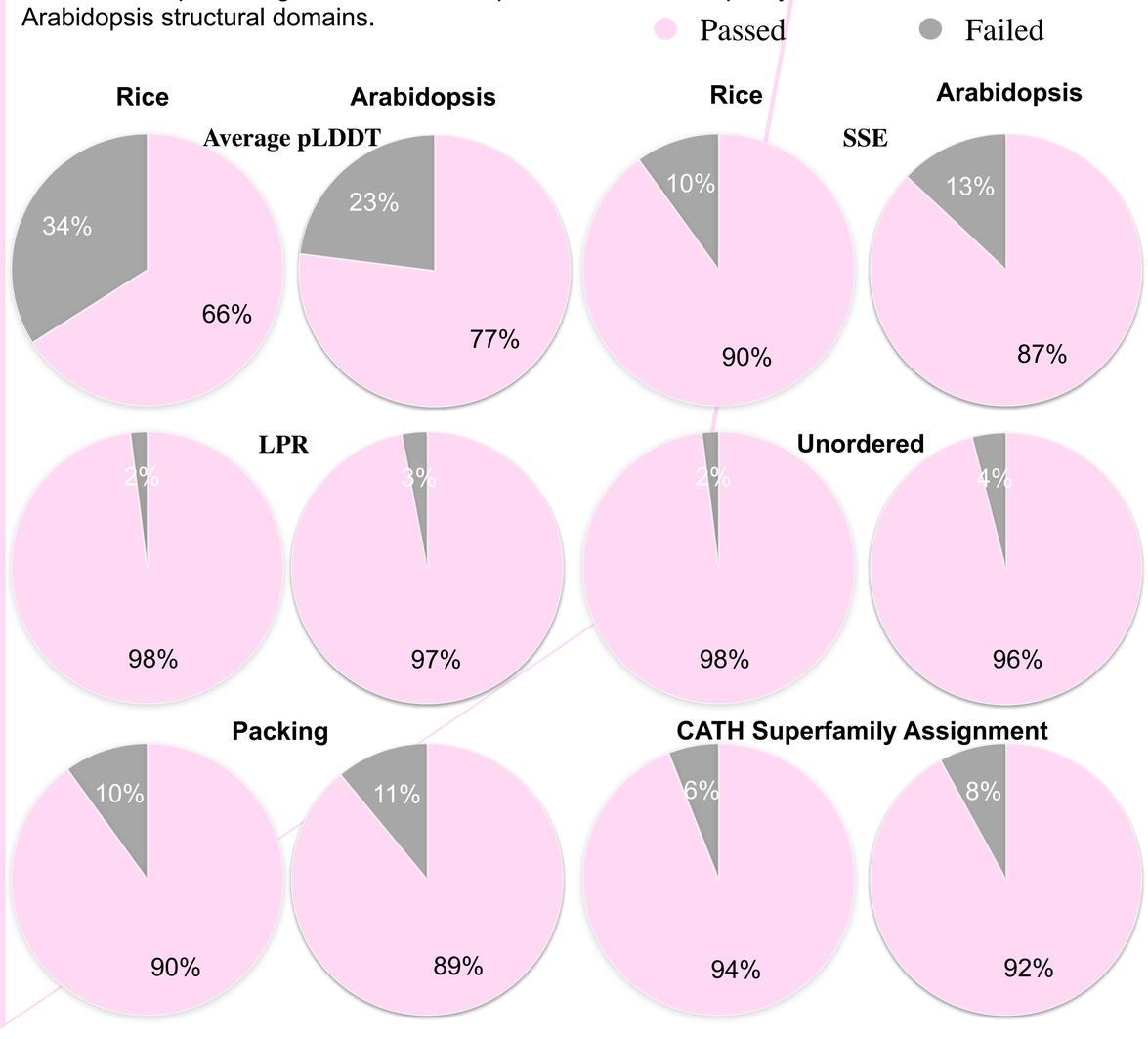
pLDDT Distribution

Our analysis shows that **a large percentage of domains lie in the highly accurate (>90) and well modelled (70-90) range** while a few are acceptable (50-70) and poor (below 50).

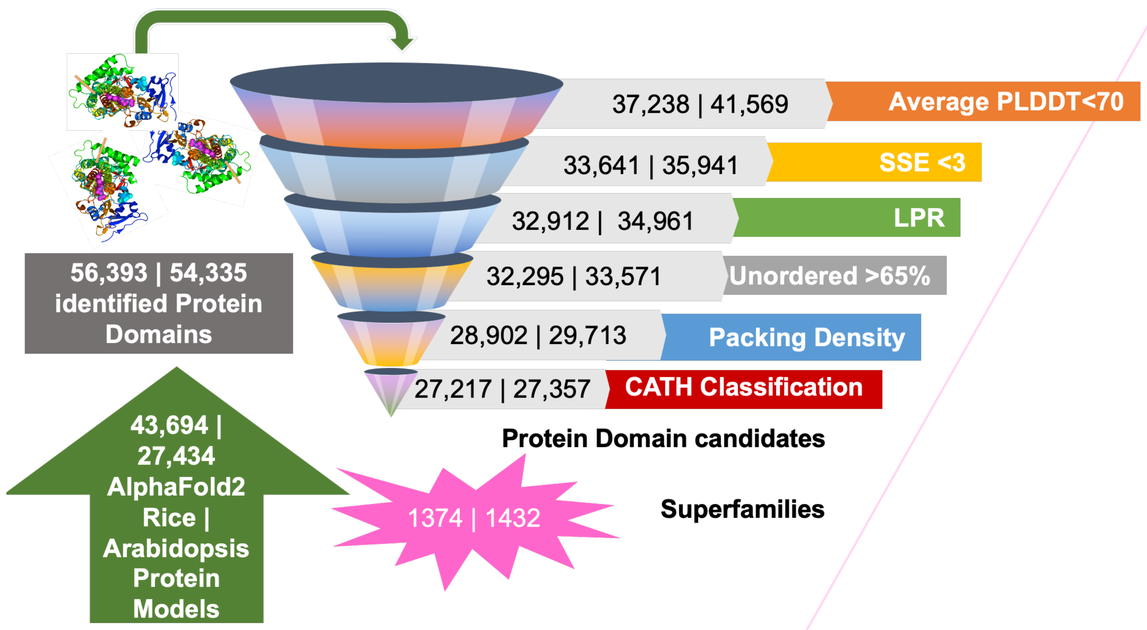


Model Quality Assessment

We show the percentage of domains that passed/failed each quality assessment filter for Rice and Arabidopsis structural domains.



Number of Domains Left After Each Quality Assessment Filter

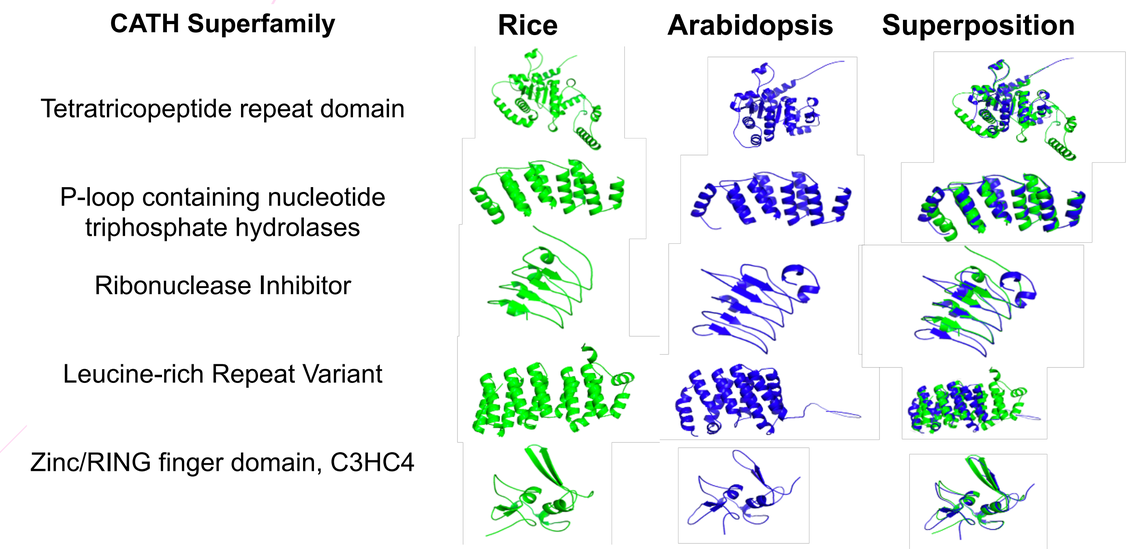


Common Superfamilies Involved in Stress Modulation

Rice and Arabidopsis domains can be mapped to 1298 common and 78|134 unique superfamilies, respectively. Table below highlights the most populated, common superfamilies between the two organisms that are involved in stress modulation.

CATH Superfamily ID	CATH Superfamily Name	Domain % in Rice	Domain % in Arabidopsis	Stress Regulation Type
1.25.40.10	Tetratricopeptide repeat domain	6.9%	8.3%	Abiotic (Oxidative, Cold)[5]
3.40.50.300	P-loop containing nucleotide triphosphate hydrolases	4.3%	4.0%	Biotic (Pathogens) , Reactive Oxygen Species[6]
3.80.10.10	Ribonuclease Inhibitor	6.7%	3.9%	Biotic (Viral), Abiotic (Salt, Cold)[7]
1.25.10.10	Leucine-rich Repeat Variant	1.7%	2.0%	Biotic (Pathogens)[8]
3.30.40.10	Zinc/RING finger domain, C3HC4 (zinc finger)	1.2%	1.8%	Abiotic (Salt, Cold)[9]

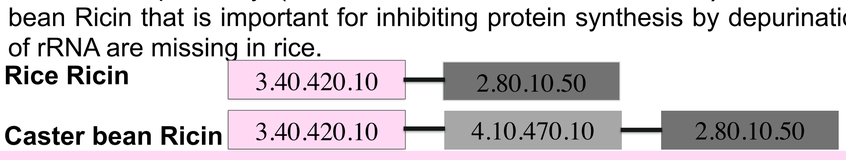
Domains representatives from both organisms with similar sequence lengths appeared to be structurally conserved in the same superfamilies.



An Interesting Rice Superfamily

Rice has 22 domain entries of **Ricin Subunit A1 superfamily (3.40.420.10)**, but Arabidopsis does not. This superfamily is involved in the production of a biotoxin in caster bean. Rice Ricin A1 domains have conserved active site residues with Ricin bio-toxin from caster bean[10]. However, superfamily (**4.10.470.10 i.e. Ricin Subunit A2)** from caster bean Ricin that is important for inhibiting protein synthesis by depurination of rRNA are missing in rice.

Conserved active site in rice



Conclusion

- **48% and 50%** of Rice and Arabidopsis domains passed the quality assessment filters, respectively. Previously present 360 and 3635 experimentally verified CATH domains from Rice and Arabidopsis have increased to **27217 and 27357** now. The structural superfamilies have spiked from 56 and 306 to **1374 and 1432**.
- The densely populated common superfamilies tend to **have a role in biotic and abiotic stress management.**
- **Rice Ricin subunit A1 superfamily** shows **sequence and structural conservation** to a domain in caster bean which is associated with a biotoxin. However, the non-toxicity of rice towards humans may be attributed to the lack of other domains found in the castor bean protein.

References

1)Khan et al. Plant Growth Regul. 90.2 (2020): 189-203.
2)Jumper et al. Nature. 596.7873 (2021): 583-589.
3)Taylor et al. J. Mol. Biol. 208.1 (1989): 1-22
4)Van Kempen et al. bioRxiv (2022).
5)Main et al. Proc. Natl. Acad. Sci. 102.16 (2005): 5721-5726.
6)Li et al. Plant Signal. Behav. (2022):2025678.
7)Holt et al. Curr. Opin. Immunol. 15.1(2003): 20-5.
8)Chisholm et al. Cell. 124.4(2006): 803-14.
9)Han et al. Int. J. Mol. Sci. 22.15(2021): 8327.
10)Polito et al. Toxins. 11.6(2019): 324.